



ForgeGIS™

GPU-Accelerated Geospatial Compute for the Agent Era

A GPU-accelerated geospatial library with a native Model Context Protocol server — as of May 2026, the only commercially-supported GPU-accelerated geospatial MCP server we have been able to identify. 239 operations spanning the working surface of professional GIS analysis, distributed as pure-Java Maven artifacts with no first-party native code. Built from the start for both programmatic pipelines and AI-agent-driven workflows — and fast where it counts: multi-stage GPU-resident pipelines run an order of magnitude ahead of equivalent GDAL CLI chains.

7.5×

**geometric-mean speedup vs.
GDAL CLI**

*no category excluded; 225 head-to-head records,
June 2026 validation run*

What it is

ForgeGIS is the GPU-accelerated compute substrate for geospatial analysis, distributed as pure-Java Maven artifacts with no first-party native code and no GDAL native bindings. It exposes 239 operations across terrain, bathymetry, hydrology, RF/visibility, spectral and ML, spatial geometry, filtering, raster I/O, routing, point cloud, and temporal analysis — callable directly from Java for programmatic pipelines, and exposed to AI agents through a dual-surface MCP server.

28×

**multi-stage GPU-resident
pipeline speedup**

*category geomean; warp & algebra pipelines
reach 36–65× at 4096²*

Why it matters

Today's geospatial stack was built for a human analyst at a desktop. Agents need different primitives — composable, callable, and fast enough that a multi-step pipeline doesn't time out. ForgeGIS provides those primitives without forcing organizations to choose between agentic workflows and the deterministic, scripted control their environments require. The numbers a scripted run depends on are identical when an agent invokes the same operation.

239

**user-facing operations across
11 categories**

*237 agent-accessible via MCP; 152 on the
pipeline registry*

How it's built differently

Data stays GPU-resident across raster operations rather than round-tripping through host memory or disk between stages — the structural reason multi-step analysis runs at machine speed. New in 1.1: a non-linear DAG pipeline engine imports existing QGIS models and runs raster and vector stages in one graph. The deployment artifact is a single shaded Java jar — no separate native install step, no GDAL system install.

WHO FORGEGIS IS FOR

Defense, intelligence, and aerospace contractors

A pure-Java Maven distribution with no first-party native code and no GDAL native bindings means ForgeGIS drops into existing JVM environments — including air-gapped and classified deployments — without coordinating separate native library distribution or shared-library signing for ForgeGIS itself. Numerical outputs are verified deterministic across surfaces, preserving audit and reproducibility guarantees when operations are invoked by AI agents.

Builders of AI agents for geospatial workflows

The dual-surface MCP server exposes a pipeline DSL that composes operations into multi-step workflows agents construct dynamically, plus 60 curated intent tools for typed single-call analytical tasks. A profile-gated tool surface keeps the default agent footprint small — agents reach professional-grade GIS analysis through MCP without needing to know geospatial internals.

Mid-market geospatial SaaS, climate, and agtech platforms

GPU-accelerated geospatial compute at server-class performance on commodity laptop hardware — the entire June 2026 validation run was measured on a single laptop with an RTX 5070. Drop ForgeGIS into existing Java services without replatforming on CUDA or coordinating native binary distribution. Existing QGIS models import and run unchanged through the new DAG pipeline engine.

PERFORMANCE HIGHLIGHTS

7.5×

geometric mean across 225 head-to-head records vs. GDAL CLI (206 wins; no category excluded; warm-mean times, June 2026 run)

9-65×

multi-stage GPU-resident pipelines at 1024²–4096² rasters (terrain, warp, algebra, and DAG chains)

3-15×

per-category geomeans across transform, resample, vectorize, zonal/focal, warp, terrain, and I/O

New in 1.1 — non-linear pipelines and a bigger vector surface. A DAG pipeline engine runs branching, non-linear workflows and imports QGIS .model3 models directly, closing the full raster → vector → raster handoff grid. The vector surface more than doubled (boolean overlay, dissolve, Voronoi, variable buffer, field calculators, and a clean-room expression interpreter), and a bundled EGM96 geoid makes vertical-datum conversion work out of the box.

WHAT'S IN THE CATALOG

Terrain · Bathymetry & Ocean · Hydrology · Visibility & RF · Spectral & ML · Spatial Geometry · Filtering, Focal, & Morphology · Raster Management, Vectorize, & I/O · Routing · Point Cloud · Temporal

239 operations total. 237 accessible via the MCP server (68 tools — 60 intent, 5 primitive handle/session, 3 pipeline-escape), 152 of them on the pipeline registry. The complete enumerated catalog and per-operation validation oracles are documented in the full Technical Brief.

Get the full Technical Brief

A technical brief with the complete 239-operation catalog, validation methodology, per-category benchmark breakdowns, and architectural deep-dive is available on request.

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